PGR216: AI and Cybersecurity Application

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Logistic regression, KNN

Kristiania

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1. What is Classification?
2. Logistic Regression
   - Linear VS Logistic regression
   - Odds of success
3. From Sigmoid to a Working Classifier
4. Visual Intuition (Simple and Visual)

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What Is a Classification Problem?

Goal: Assign an input to one of **discrete categories (classes)**.

Input: feature vector

$$x = [x_1, x_2, \ldots, x_d]$$

Output: class label from a finite set

$$y \in \{\text{class 1}, \text{class 2}, \ldots\}$$

Examples:

- Spam detection: $y \in \{0, 1\}$
- Disease diagnosis: $y \in \{\text{healthy}, \text{diseased}\}$
- Sentiment analysis: $y \in \{\text{positive}, \text{neutral}, \text{negative}\}$

We want to learn a function:

$$f(x) \to y$$

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Types of Classification Problems

1) Binary Classification

$$y \in \{0, 1\}$$

Example: spam vs not spam

2) Multiclass Classification

$$y \in \{1, 2, \ldots, K\}$$

Example: digit recognition (0–9)

3) Multilabel Classification

$$y = [y_1, y_2, \ldots, y_K], \quad y_k \in \{0, 1\}$$

Example: an image may contain *cat*, *dog*, and *car* at the same time

Common goal: Separate classes in the feature space using decision boundaries.

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Classification as Supervised Learning

Classification belongs to **supervised learning**.

We are given a training dataset:

$$\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$$

- x_i : input features
- y_i : known correct label (provided by humans / experts)

The model learns from these labeled examples to predict labels for new, unseen data.

Learn from examples $\Rightarrow$ Predict new labels

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Classification

A machine learning problem is defined as a classification problem when the independent variable/variables are continuous in nature and dependent variable is discrete. A couple of classification examples from the real world are:

- Classifying a cyber attack (Intrusion, Malware, DDoS) from network flow data
- Classifying a virus as Covid-19 or the normal flue
- Checking the authenticity of a transaction
- Classifying spam email from the normal emails.

In a nutshell, logistic regression is used when a dataset with one or more than one independent variables is used to determine the binary output of the dependent variable.

Car Breakdown

Car Breakdown

You bought an old car and it is approved until the Jan 2021.

- You didn't perform any service since you bought the car
- Now you wonder, how long will it take to breakdown.
- On other words, what is the probability that the car will breakdown before Jan 2021.

Courtesy:https://www.arag.co.uk/broker-products/assistance/motor-breakdown-solutions/

Machine learning Exam

Machine learning Exam

You took a course in machine learning.

- The course has a couple of compulsory assignments.
- You submitted all the assignments but some of them were approved and some was not approved.
- What is the probability that you will pass the course.

Courtesy:https://favpng.com/png_view/student-homework-test-education-student-png/BSkm3RxV

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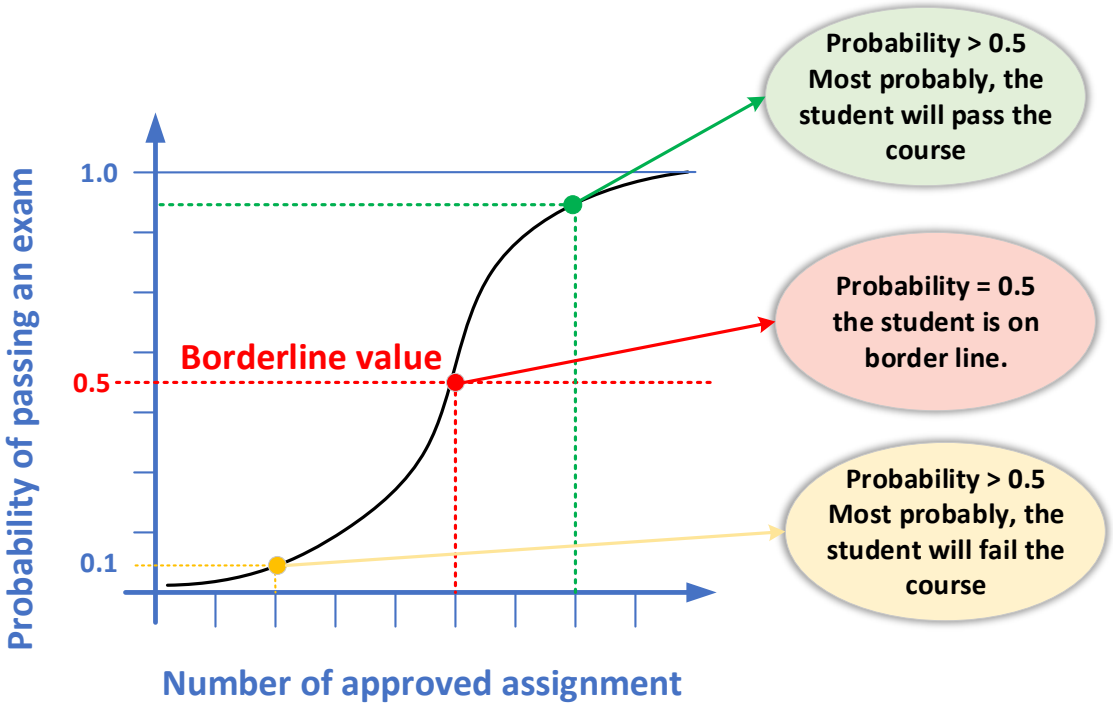
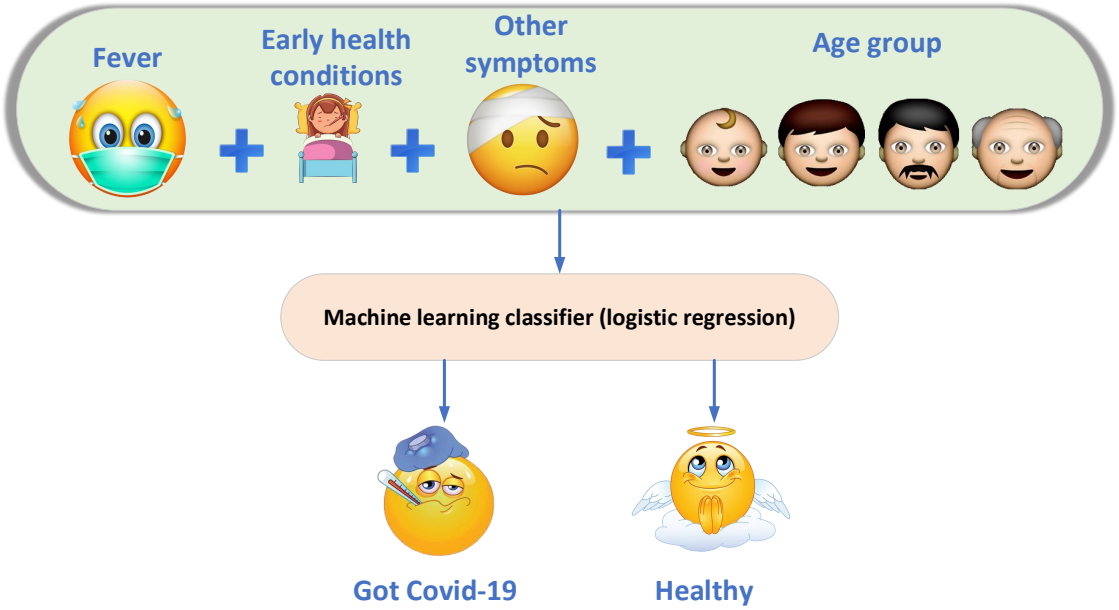


Covid-19

- You got a flu and other symptoms
- You have a health record
- You have other allergies
- Considering your age and all the other factors, what is the chance that you got Covid-19



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Problem Statement

If we take the example of email classification, we may define the task as:
For every incoming email, decide:

- $1 -> \text{Spam}$
- $0 -> \text{Not spam}$

We don't look at the email manually. We extract features from it.

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Problem Statement

- Features could be Numeric.
- Feature could be categorical.

Example features (inputs x):

- x_1: Number of links in the email
- x_2: Length of the email (in words)
- x_3: Contains words like "free", "win", "offer" (0 or 1)
- x_4: Contains urgency keywords (limited time offer, click the link)

So each email becomes a vector:

$$*x = [x_1, x_2, x_2, x_4] \tag{1}$$

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Feature Vector

An email is represented as a vector:

$$x = [x_1, x_2, x_3]$$

Example:

- Email A: 10 links, 200 words, suspicious words present

$$x^{(A)} = [10, 200, 1]$$

- Email B: 0 links, 20 words, no suspicious words

$$x^{(B)} = [0, 20, 0]$$

Goal: Use x to predict a label $y \in \{0, 1\}$.

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What Should the Model Output?

For each email, the true label is:

$$y = \begin{cases} 1 & \text{Spam} \\ 0 & \text{Not spam} \end{cases}$$

But we often want more than a hard label:

- Email A: $P(\text{spam} \mid x) = 0.95$ (almost certainly spam)
- Email B: $P(\text{spam} \mid x) = 0.55$ (uncertain / suspicious)
- Email C: $P(\text{spam} \mid x) = 0.02$ (very likely safe)

So the model output should satisfy:

$$0 \leq \text{output} \leq 1$$

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First Idea: Can We Use Linear Regression?

You may think:

"We already learned linear regression. . . can we just use that?"

Linear regression model:

$$\hat{y} = w^T x + b$$

Question: If $y \in \{0, 1\}$, does it make sense for $\hat{y}$ to be any real number?

What Happens If We Try Linear Regression?

Suppose the model learns:

$$\hat{y} = 0.8x_1 + 0.01x_2 + 2x_3 - 3$$

Test on an email with:

$$x = [10, 200, 1]$$

Then:

$$\hat{y} = 0.8(10) + 0.01(200) + 2(1) - 3 = 9$$

Prediction = 9 (What does "9" mean for a binary label?)

Even Worse: Linear Regression Can Output Negatives

Test on another email:

$$x = [0, 20, 0]$$

Then:

$$\hat{y} = 0.8(0) + 0.01(20) + 2(0) - 3 = -2.8$$

Prediction = -2.8 (What does "negative spam" mean?)

Core Problem with Linear Regression for Classification

Linear regression outputs:

$$\hat{y} \in (-\infty, +\infty)$$

But for binary classification, we need:

$$\hat{y} \in [0, 1]$$

So what is wrong?

- The **linear part** $w^T x + b$ is useful (it can form a decision boundary)
- The **output range** is the problem

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Keep the Linear Model, Change the Output

We like this score:

$$z = w^T x + b$$

because it measures how "spam-like" an email is.

But we need a function that converts z into a probability:

- Squashes any real number to the range $[0, 1]$
- Can be interpreted as $P(y = 1 \mid x)$

This is exactly what the logistic function does.

One-Sentence Take away

Logistic regression = linear model + logistic output

- Linear regression: $\hat{y} = w^T x + b$ (unbounded)
- Logistic regression: $\hat{p} = \sigma(w^T x + b)$ (bounded in $[0, 1]$)

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Logistic Regression

Logistic regression is a statistical analysis algorithm which is used when the dependent variable is dichotomous (discrete and binary) in nature. In machine learning, it is one of the basic, simple, and popular algorithm to solve the classification problems.

Logistic

The name "Logistic Regression" comes from the fact that the underlying principle are quite similar to Linear Regression. Similarly, the name "Logistic" comes from the Logit function that it use for defining the classification boundaries.

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Linear VS Logistic regression

Linear regression

- Use to solve regression problems
- The dependent variable is continuous
- Helps in estimating the dependent variable when there is change in the independent variable
- It is a straight line

Logistic regression

- Use to solve classification problems
- The dependent variable is discrete
- Helps in calculating the possibility of a particular event taking place
- An S-shaped curve named as sigmoid

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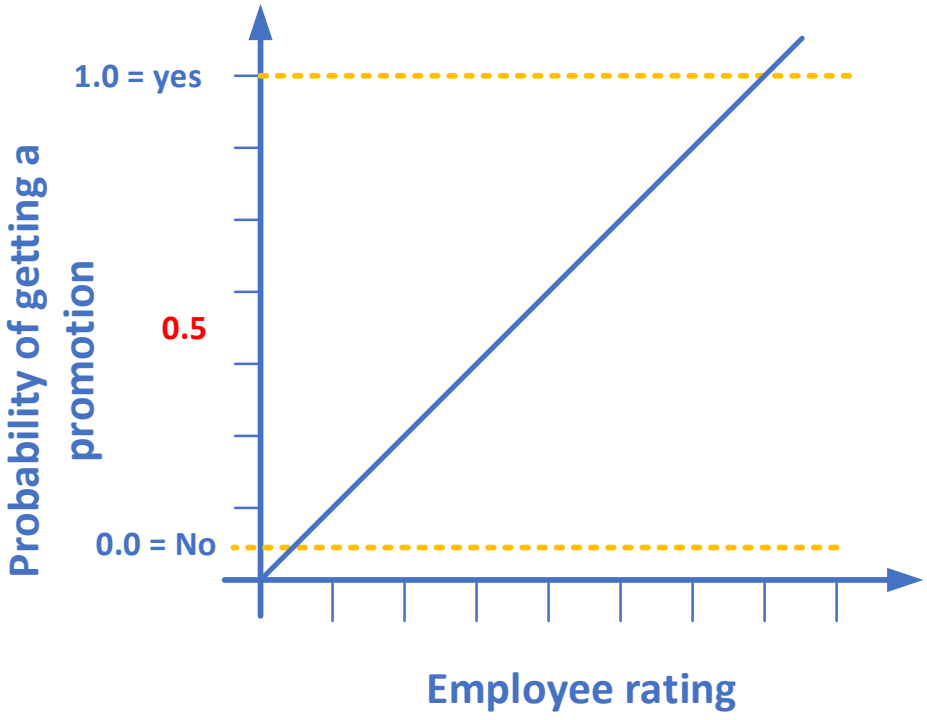
Linear regression model

Let's find the probability of an employee promotion based on it's rating.

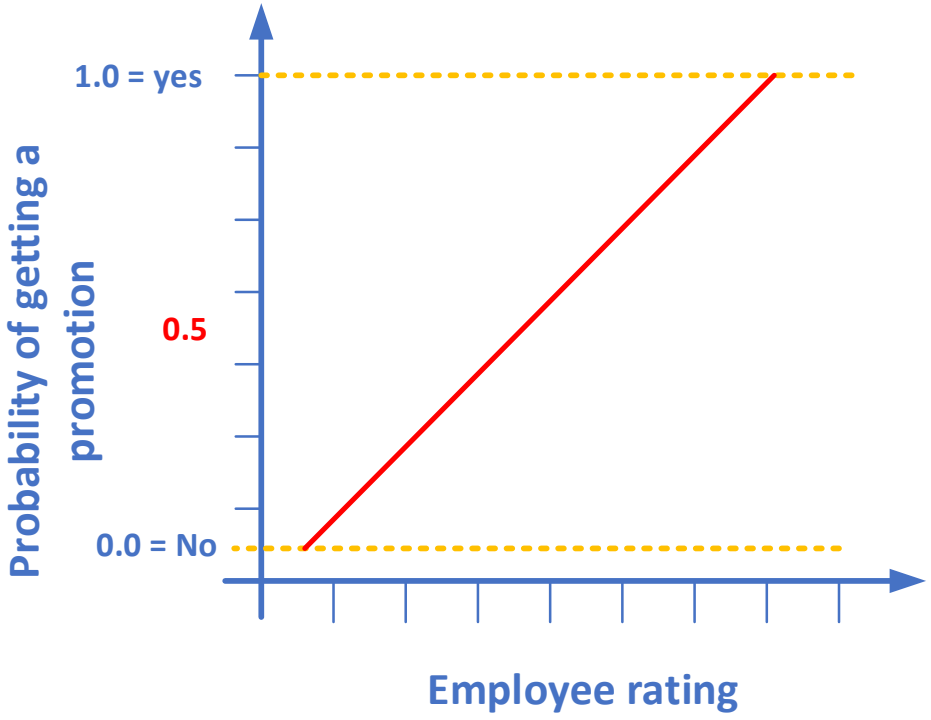
- Rating could be achieved from productivity
- The feedback of clients
- Other parameters define by the employer.

If we want to solve this problem through linear regression, the model will look like a linear line.

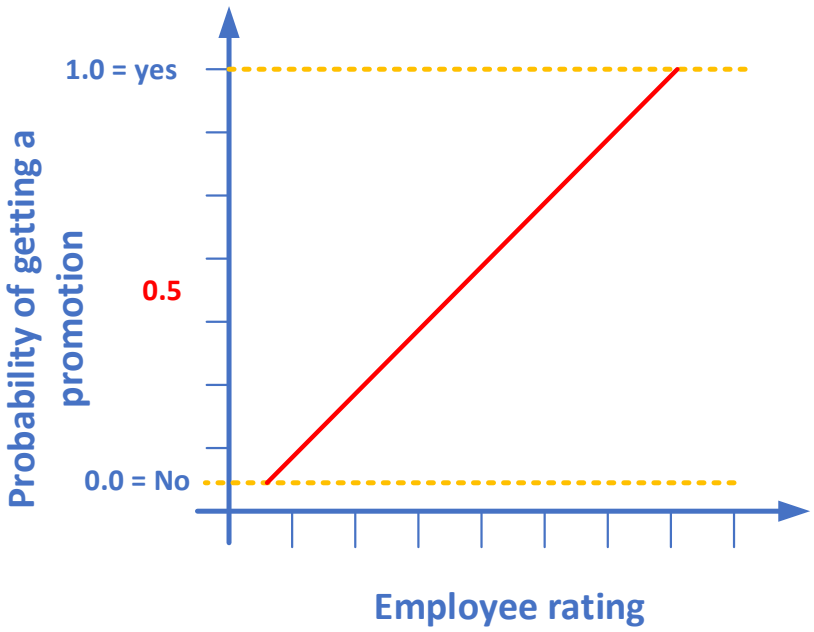


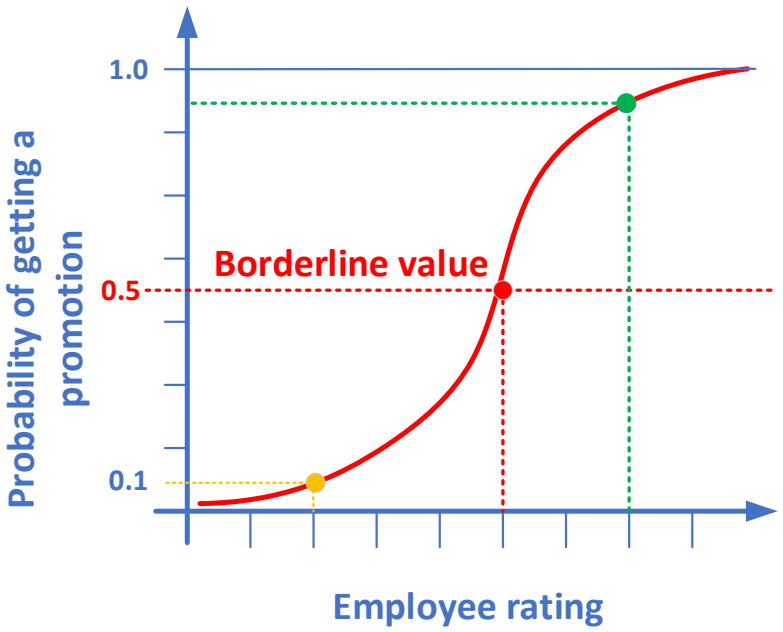
Linear regression Model

- The values of linear model stretches for $-\infty$ to $+\infty$.
- For classification, we are only interested in values between 0 and 1.



Linear VS Logistic regression

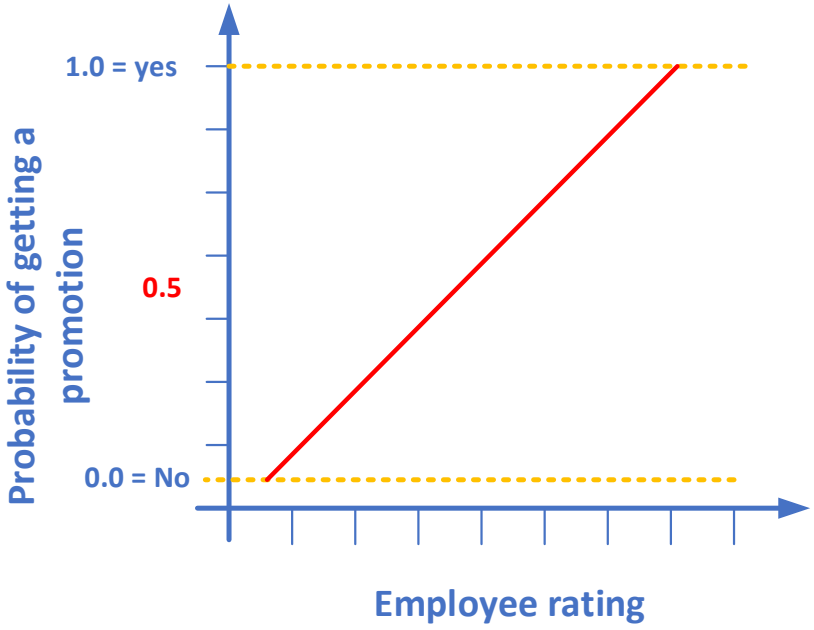


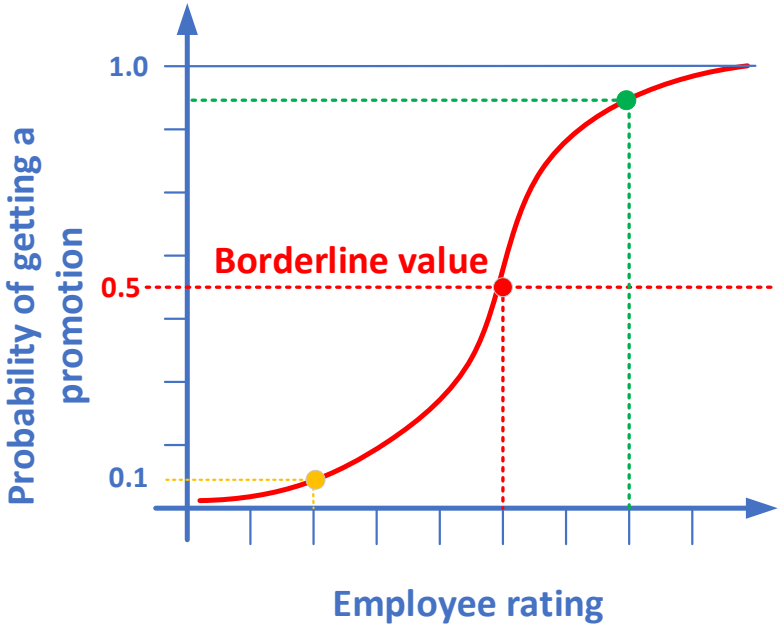


So, in the left graph, a typical linear regression model is depicted. While in the right graph, logistic regression is shown. It can be clearly seen that the linear regression model gives equal weightage to all the values. However, in logistic regression, once the borderline is crossed in any direction, it is more probable to get either of the class.

Linear VS Logistic regression

Mathematically, How can we move from a linear regression model to a logistic regression?





Odds of success

The mathematical foundation of logistic regression lies in understanding the probability concept of odds of success. It is defined as:

$$Odds(\theta) = \frac{\text{Probability of an event happening}}{\text{Probability of an event not happening}} \tag{2}$$

$$Odds(\theta) = \frac{p}{1-p} \tag{3}$$

Where p is representing the probability of an event. But generally, p could represent the probability of any favorable output by the model.

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Odds of success

It is important to note that the value of odds of success ranges from 0 to ∞. While p is always between 0 and 1.

$$Odds(\theta) = \frac{p}{1-p}$$

Linear regression model

From the linear regression model, we know the model is represented by equation of a straight line:

$$Y = \beta_0 + \beta_1 x$$

Where

- β_0 is the y-intercept
- β_1 is the slope of the line.
- x is the independent and y is the dependable variable.

Odds of success

Linear regression model

$$Y = \beta_0 + \beta_1 x$$

Model for the classification

$$Y = \left(\frac{p(x)}{1 - p(x)} \right)$$

By equating the log of classification model with the linear regression model,

$$\log \left(\frac{p(x)}{1 - p(x)} \right) = \beta_0 + \beta_1 x \tag{4}$$

Odds of success

By taking exponential on both sides of the equations:

$$\exp^{\log}\left(\frac{p(x)}{1-p(x)}\right) = \exp^{\beta_0+\beta_1 x} \tag{5}$$

log is canceled by the exp and the expression reduces to:

$$\left(\frac{p(x)}{1-p(x)}\right) = \exp^{\beta_0+\beta_1 x} \tag{6}$$

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Odds of success

Through substitution, we can get:

$$Y = \exp^{\beta_0 + \beta_1 x}$$

Then

$$\left(\frac{p(x)}{1 - p(x)}\right) = Y$$

$$p(x) = Y(1 - p(x))$$

$$p(x) = Y - Yp(x)$$

$$p(x) + Yp(x) = Y$$

$$p(x)(1 + Y) = Y$$

Odds of success

$$p(x)(1+Y) = Y$$

$$p(x) = \frac{Y}{(1+Y)}$$

$$p(x) = \frac{\exp^{\beta_0+\beta_1 x}}{(1+\exp^{\beta_0+\beta_1 x})}$$

Equation of Sigmoid function

$$p(x) = \frac{\exp^{\beta_0+\beta_1 x}}{(1+\exp^{\beta_0+\beta_1 x})}$$

$$p(x) = \frac{1}{(1+\exp^{-(\beta_0+\beta_1 x)})}$$

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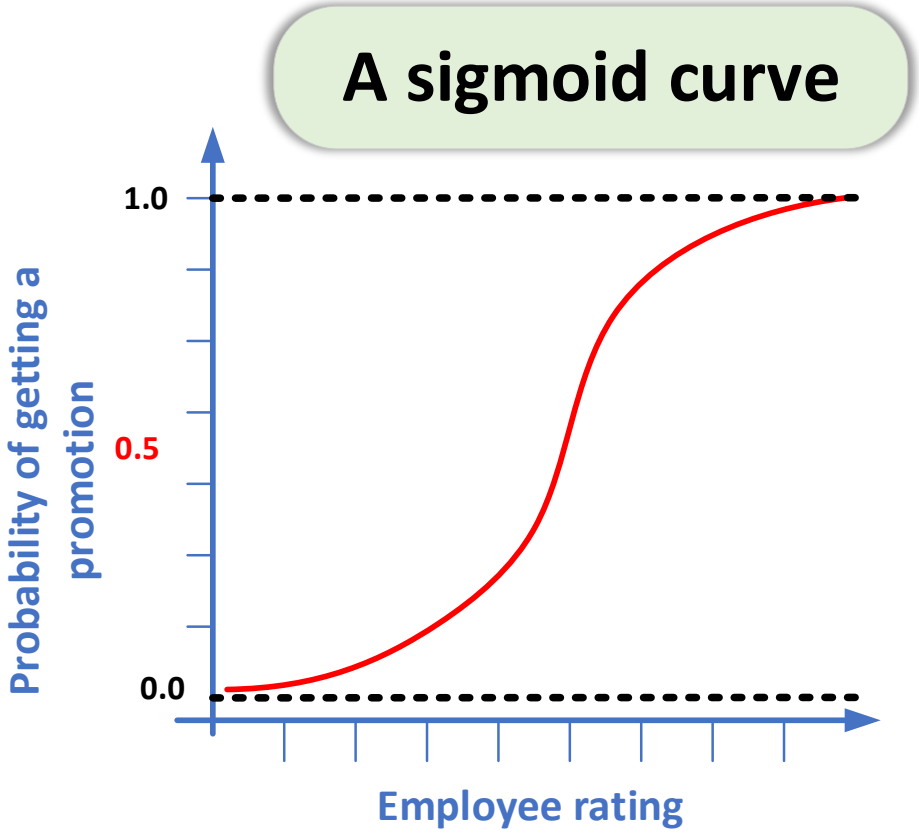
Sigmoid function

Sigmoid is a standard mathematical continuous differential function. It is commonly used in neural networks as the activation function and has a characteristic "S" shaped curve. The domain of the sigmoid function ranges from $-\infty$ to ∞ while the range is always between zero and one. This property makes it extremely useful for classification problems.



Logistic Regression

The logistic function is a sigmoid function, which takes any real numbers as input (features), and outputs a value between zero and one:

$$\text{Logistic function} = \frac{1}{1 + \exp^{-arg}} \tag{7}$$

For the logistic regression, the argument is the combination of the features extracted from the data.

$$p(x) = Y = \frac{1}{1 + \exp^{-(\beta_0 + x\beta_1)}} \tag{8}$$

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Loss function

As the loss function for linear regression was a squared loss. The commonly used loss function for logistic regression is called the cross entropy loss or the Log Loss and it is define as:

$$\text{loss} = \sum_{x,Y \in D} -Ylog(Y^{'}) - (1 - Y)log(1 - Y^{'}) \tag{9}$$

Cost Function

$$\text{loss} = \sum_{x,Y \in D} -Ylog(Y^{'}) - (1-Y)log(1-Y^{'}) \tag{10}$$

if Y = 1:

$$-Ylog(Y^{'}) \tag{11}$$

if Y = 0:

$$-(1-Y)log(1-Y^{'}) \tag{12}$$

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Loss function

$$\text{loss} = \sum_{x,Y \in D} -Ylog(Y^{'}) - (1-Y)log(1-Y^{'}) \tag{13}$$

- Given the fully supervised nature of the problem, $(x, Y \in D)$ is the labeled dataset where (x, y) shows one label pair i.e. pair of dependent and independent variables.
- Y is the dependent variable. For the classification, the value of Y will be either 0 or 1.
- $Y^{'}$ is the predicted value of the logistic regression which will be somewhere in between 0 and 1.

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The Sigmoid (Logistic) Function Interpretation

Define the sigmoid:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Properties:

- Output range: $\sigma(z) \in (0, 1)$
- $\sigma(0) = 0.5$
- Monotonic increasing: larger $z \Rightarrow$ larger probability

Logistic regression uses:

$$p(y = 1 \mid x) = \sigma(w^T x + b)$$

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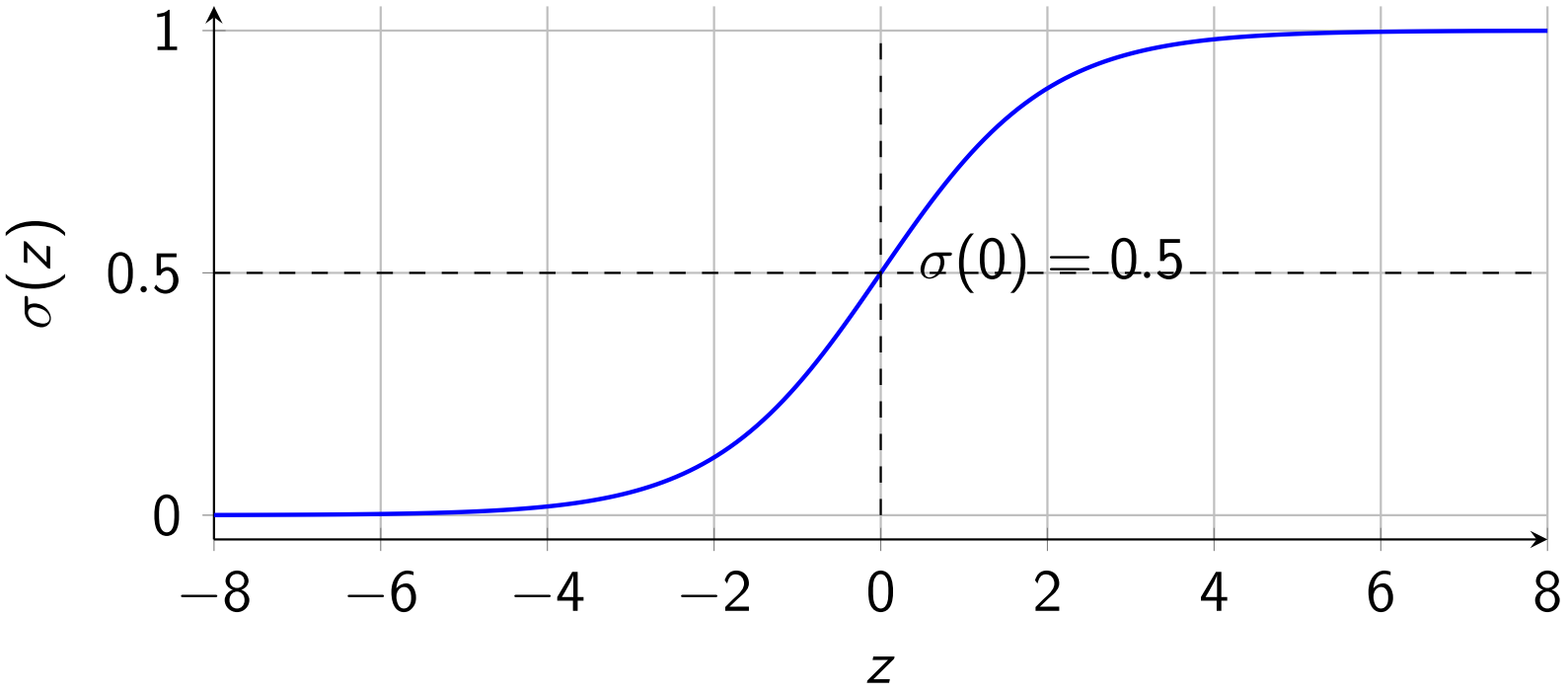
Plot: Sigmoid Function



Interpretation: z is a score; sigmoid turns it into a probability.

Logistic Regression Model (Binary Classification)

Let $x \in \mathbb{R}^d$, $w \in \mathbb{R}^d$, $b \in \mathbb{R}$.

$$z = w^T x + b \qquad p = P(y = 1 \mid x) = \sigma(z)$$

Decision rule (thresholding):

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq \tau \\ 0 & \text{otherwise} \end{cases}$$

Common default: $\tau = 0.5$ (but can be changed for imbalanced data).

Key point: The *boundary* is linear in x even though probabilities are nonlinear in z.

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Logistic Regression Model

We compute a score:

$$z = w^T x + b$$

Then convert it into probability using sigmoid:

$$p = \sigma(z)$$

$$p = P(y = 1 \mid x)$$

Interpretation:

- Large $z \Rightarrow p$ close to 1 (very likely spam)
- Small $z \Rightarrow p$ close to 0 (very likely not spam)

How Do We Make a Final Decision?

The model gives a probability p.
We choose a threshold (usually 0.5):

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq 0.5 \\ 0 & \text{if } p < 0.5 \end{cases}$$

This means:

- If the email is more than 50% likely to be spam $\Rightarrow$ mark as spam
- Otherwise $\Rightarrow$ not spam

What Is the Decision Boundary?

The boundary happens when:

$$p = 0.5$$

Sigmoid gives 0.5 when its input is 0:

$$w^T x + b = 0$$

This is a **straight line** (or plane in higher dimensions).

- One side $\Rightarrow$ spam
- Other side $\Rightarrow$ not spam

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What Should the Model Learn? (Intuition)

For each training email:

- If $y = 1$ (spam) $\Rightarrow$ we want p close to 1
- If $y = 0$ (not spam) $\Rightarrow$ we want p close to 0

So the model should be:

- Happy when predictions are correct
- Penalized when predictions are wrong

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How Do We Measure "Wrongness"? (Loss Idea)

We use loss called **cross-entropy loss**.

- If the model says $p = 0.9$ and the true label is $1 \Rightarrow$ small loss
- If the model says $p = 0.1$ and the true label is $1 \Rightarrow$ very large loss
- Wrong and confident predictions are punished a lot

This forces the model to learn good probabilities.

How Does the Model Improve?

The model starts with random w and b (or β and β_0).
Then repeatedly:

1. Make predictions on training data
2. Measure how wrong it is (loss)
3. Slightly adjust w and b (β, β_0) to reduce the loss

This process is called **gradient descent**.

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Very Important Insight

The update rule turns out to be very simple:

$$\text{If prediction is too high} \Rightarrow \text{decrease weights}$$
$$\text{If prediction is too low} \Rightarrow \text{increase weights}$$

So the model slowly moves the decision boundary to better separate the classes.

Why Logistic Regression Is Interpretable

Recall:

$$z = w^T x + b$$

Each weight tells us how important a feature is.

- Positive weight $\Rightarrow$ increases chance of spam
- Negative weight $\Rightarrow$ decreases chance of spam
- Larger magnitude $\Rightarrow$ more influence

Very useful in Cyber Security, medicine, finance, social sciences and many other domains.

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Advantages of Logistic Regression

- Simple and fast
- Works well with small datasets
- Gives probabilities (not just labels)
- Easy to interpret
- Strong baseline model

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Limitations

- Can only learn **linear** boundaries
- Struggles with complex patterns
- Needs good features

Big Picture

Logistic regression is:

$$\text{Linear model} + \text{sigmoid} + \text{smart training}$$

This idea is the foundation of neural networks.

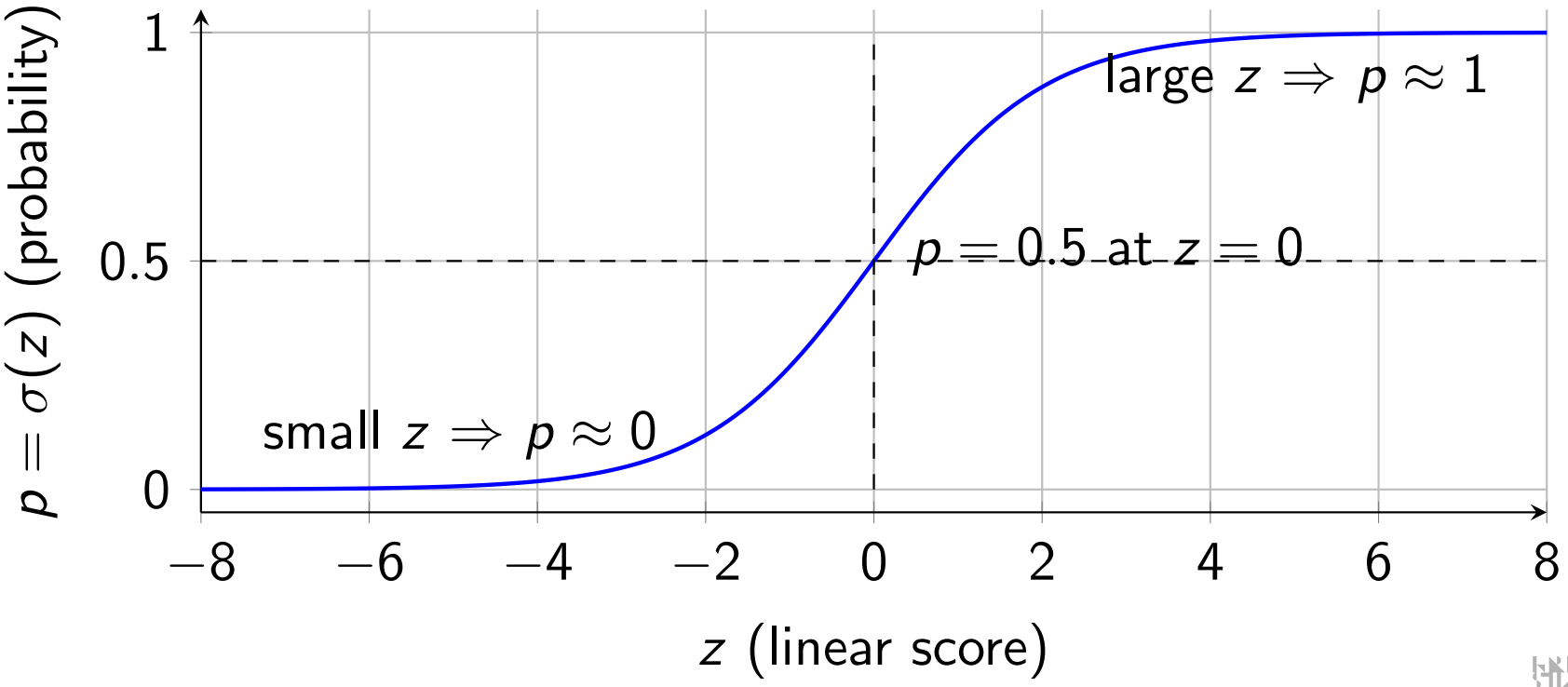
Visualization: From Score z to Probability p

We compute a score:

$$z = w^T x + b$$

and convert it to probability:

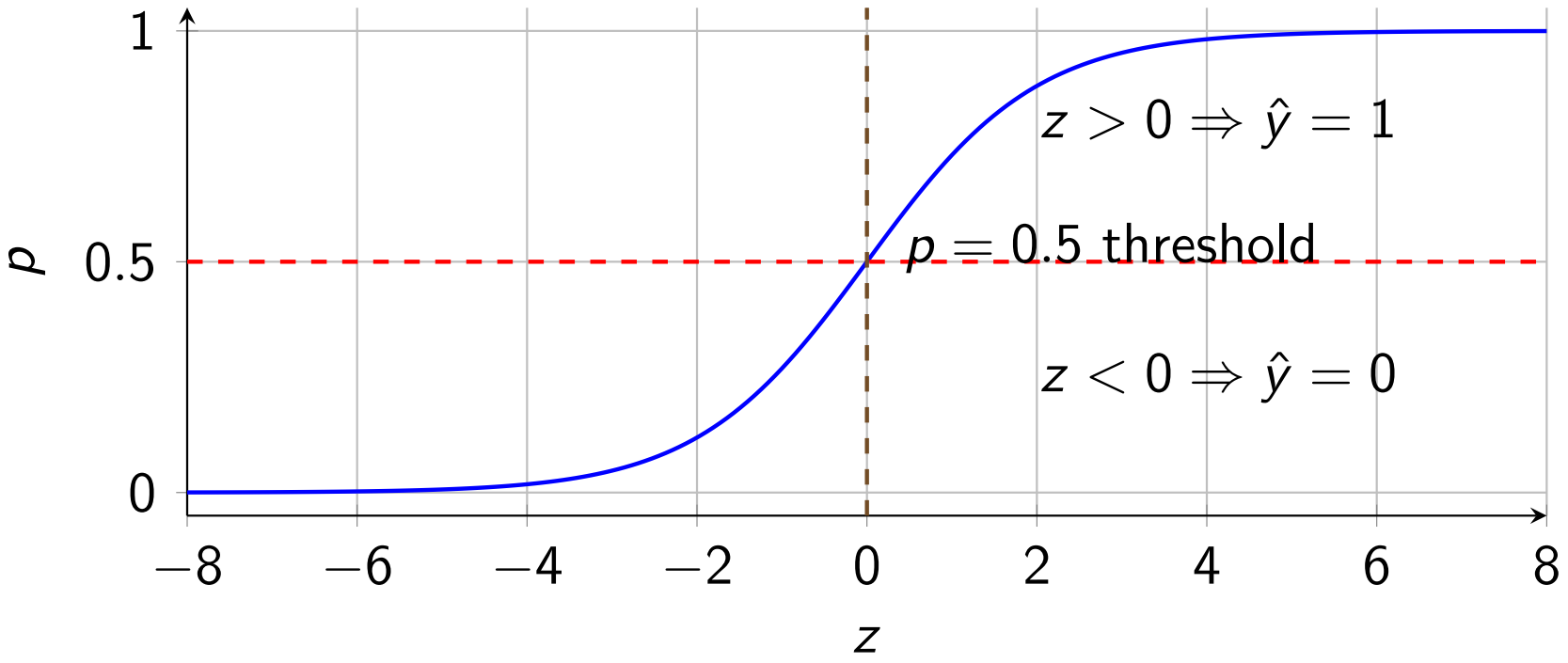
$$p = \sigma(z)$$



Visualization: Thresholding Turns Probability into a Label

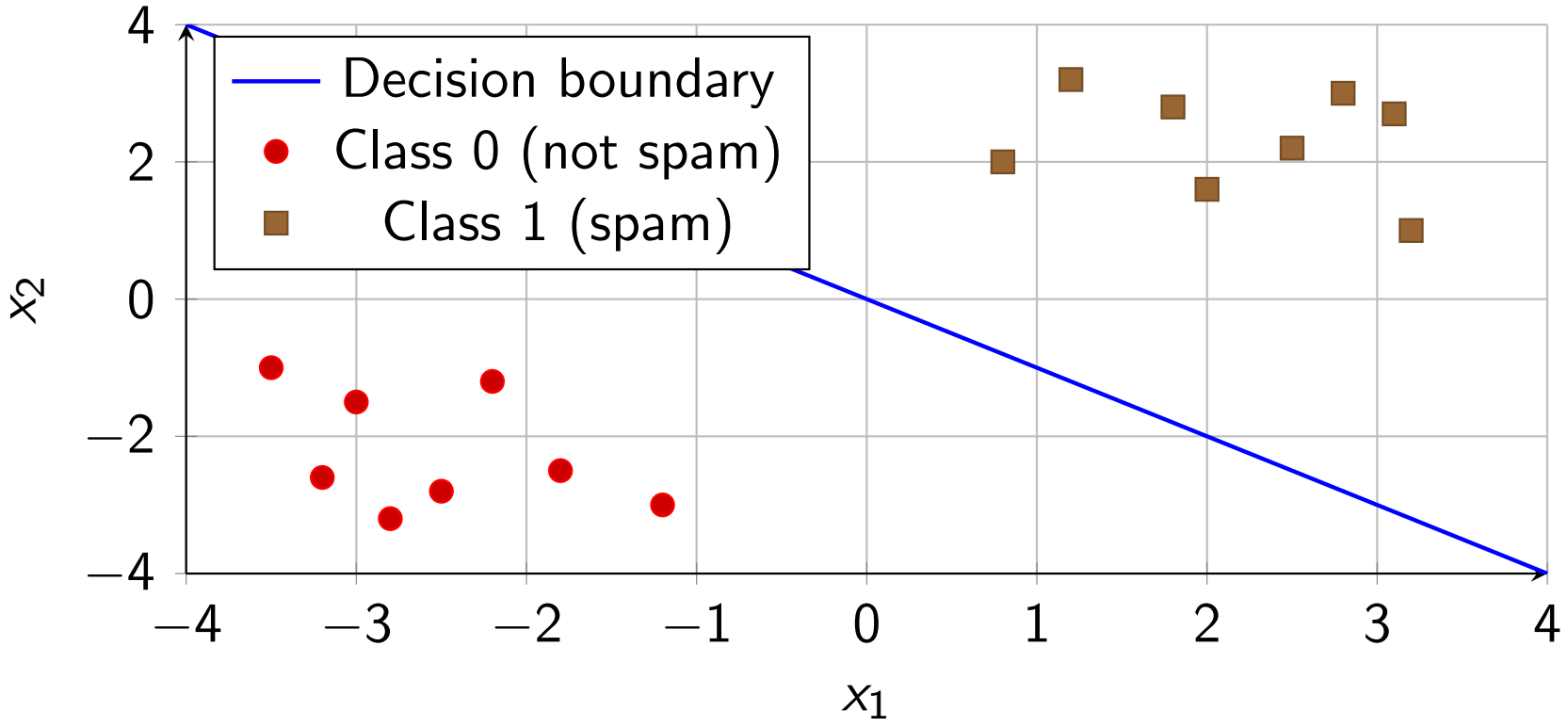
Prediction rule with threshold $\tau = 0.5$:

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq 0.5 \\ 0 & \text{if } p < 0.5 \end{cases}$$



Visualization: A Simple 2D Classification Example

Idea: points on one side of a line are class 0, the other side class 1.



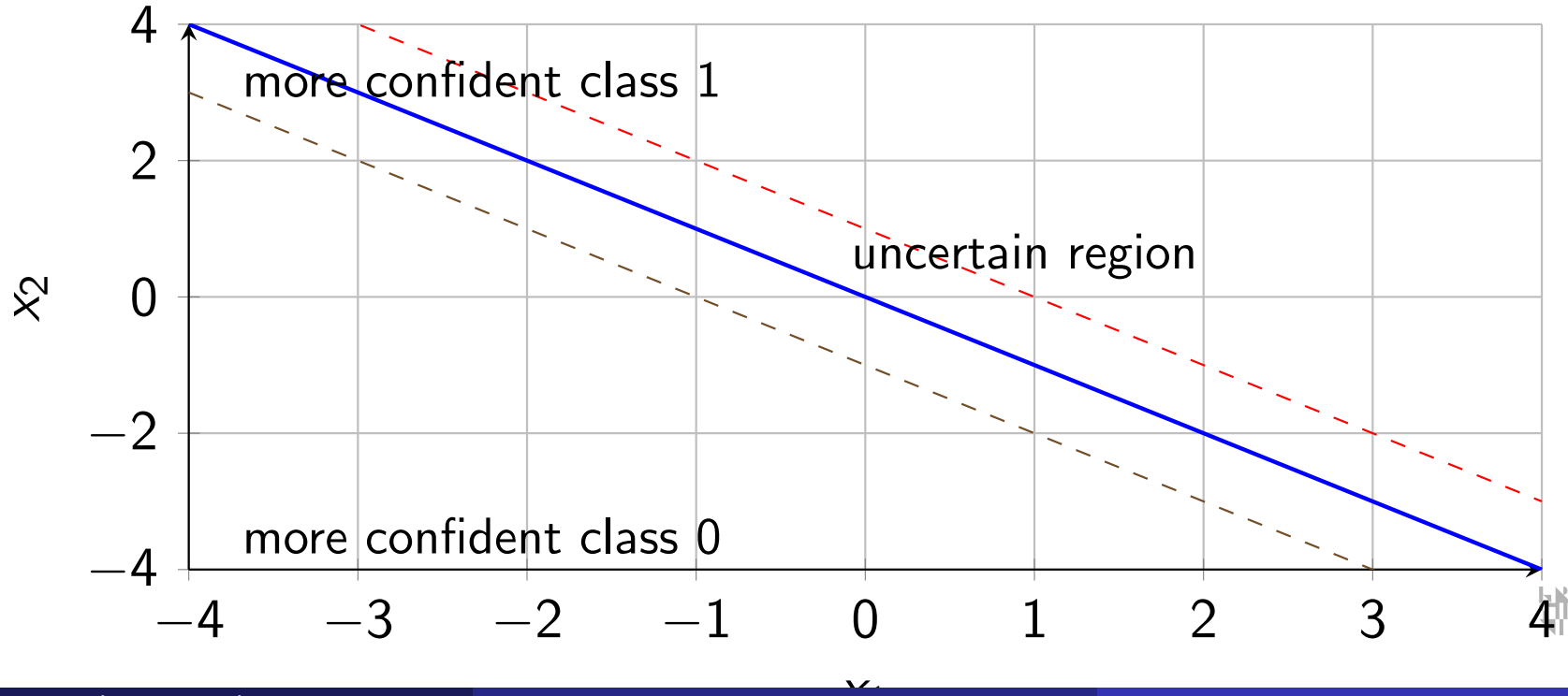
Logistic regression learns a **straight-line** boundary in feature space.

Visualization: "Confidence" Increases Away from the Boundary

Recall:

$$z = w^T x + b$$

- Near the boundary: $z \approx 0 \Rightarrow p \approx 0.5$ (uncertain)
- Far from the boundary: large $|z| \Rightarrow p$ close to 0 or 1 (confident)



Visualization: Why Accuracy Alone Can Mislead (Imbalanced Data)

Example: 1000 emails, only 50 are spam.
A dumb classifier that always predicts "not spam" gets:

$$\text{accuracy} = \frac{950}{1000} = 95\%$$

but it catches **zero** spam.

Confusion matrix idea:

	Predicted spam	Predicted not spam
Actual spam	TP	FN
Actual not spam	FP	TN

So we also look at:

- Precision: "When I say spam, how often am I right?"
- Recall: "How much spam did I catch?"

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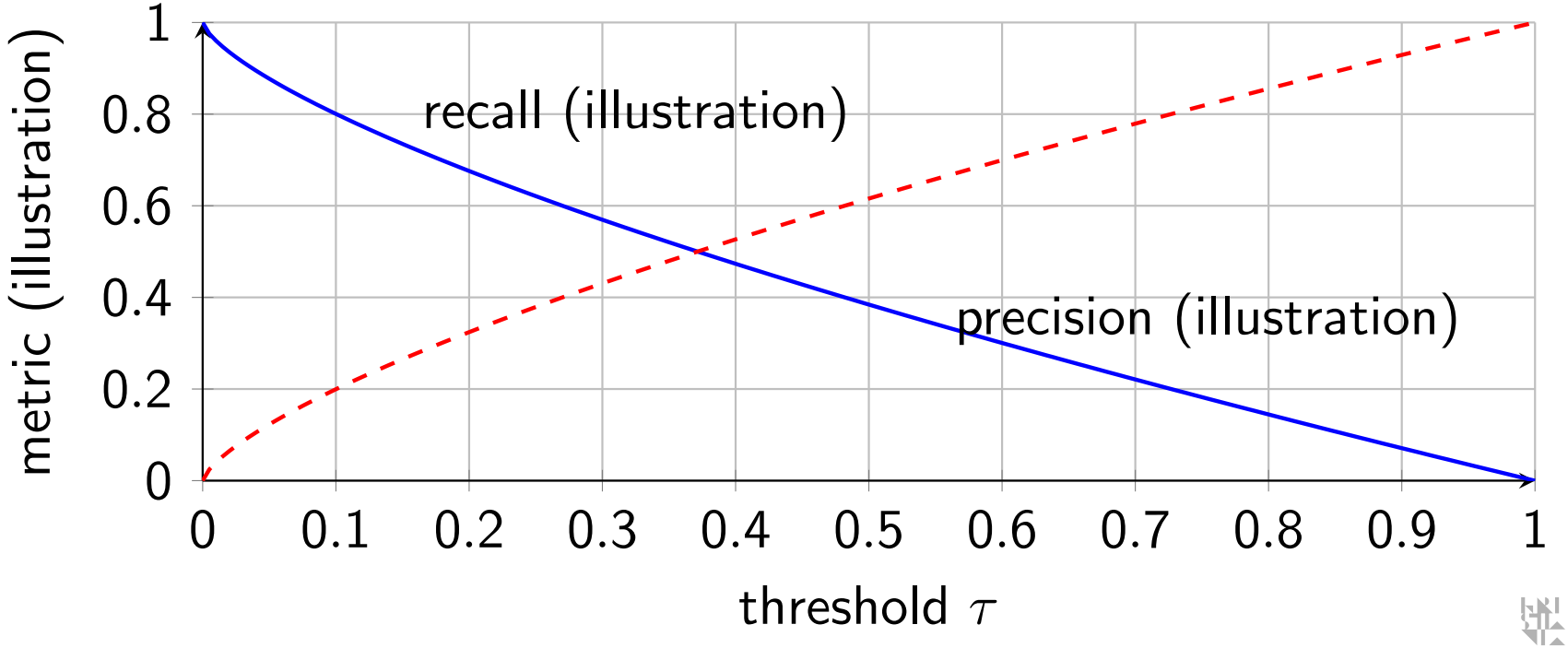
Visualization: Precision–Recall Trade-off by Changing Threshold

If we change threshold τ:

- Lower τ (e.g., 0.3): catch more spam (higher recall), but more false alarms
- Higher τ (e.g., 0.8): fewer false alarms (higher precision), but miss more spam



Mini Visual Summary

1. Compute a linear score $z = w^T x + b$
2. Use sigmoid to get probability $p = \sigma(z)$
3. Threshold p to get a class label
4. Learn w, b by reducing mistakes on training data

Linear boundary + probabilistic output

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Other Classifiers

Is there any non-parametric classifier?